VSDBabySoC SKY130 HD

POST-ROUTE STATIC TIMING ANALYSIS

16-Corner PVT Comparison and Physical-Design Evidence

Anirudh

24EG104A59

Technology
SKY130 HD standard-cell library

Tools
OpenROAD / OpenSTA

ENGINEERING REPORT • TIMING ANALYSIS • PHYSICAL-DESIGN REVIEW

Executive Summary

This report compares post-route Static Timing Analysis (STA) metrics for VSDBabySoC using the SKY130 HD standard-cell library across 16 process-voltage-temperature (PVT) corners. Setup WNS, setup TNS, hold worst slack and hold TNS are reviewed alongside physical-design evidence.

Key result: the worst setup WNS is -32.03 ns at ss_n40C_1v28, and the worst setup TNS is -15425.60 ns at the same corner. Hold worst slack is positive in all supplied rows and hold TNS is 0.00 ns throughout. STA completion does not establish timing closure.

1. Project and Methodology

The reference flow uses the routed database (5_route.odb), timing constraints (5_route.sdc), extracted parasitics (6_final.spef), SKY130 HD Liberty timing models, and macro libraries avsddac.lib and avsdpll.lib. The reference clock is **create_clock -name clk -period 11.0000 [get_pins {pll/CLK}]**, equivalent to a nominal constrained frequency of about 90.91 MHz. This is a target constraint, not proof of achieved frequency.

The values below follow the supplied results table. Verify them against the original per-corner logs before signoff. Using the same SPEF across all Liberty runs isolates timing-model differences but does not represent distinct extracted RC corners.

2. STA Metrics

Setup: verifies data arrives early enough before the capture edge. Negative setup slack indicates a failing check. Hold: verifies data does not arrive too soon after the capture edge. WNS: minimum slack among checked paths. TNS: aggregate negative slack across violating endpoints/paths. Positive hold slack and zero hold TNS do not compensate for setup violations.

3. All-Corner Timing Results

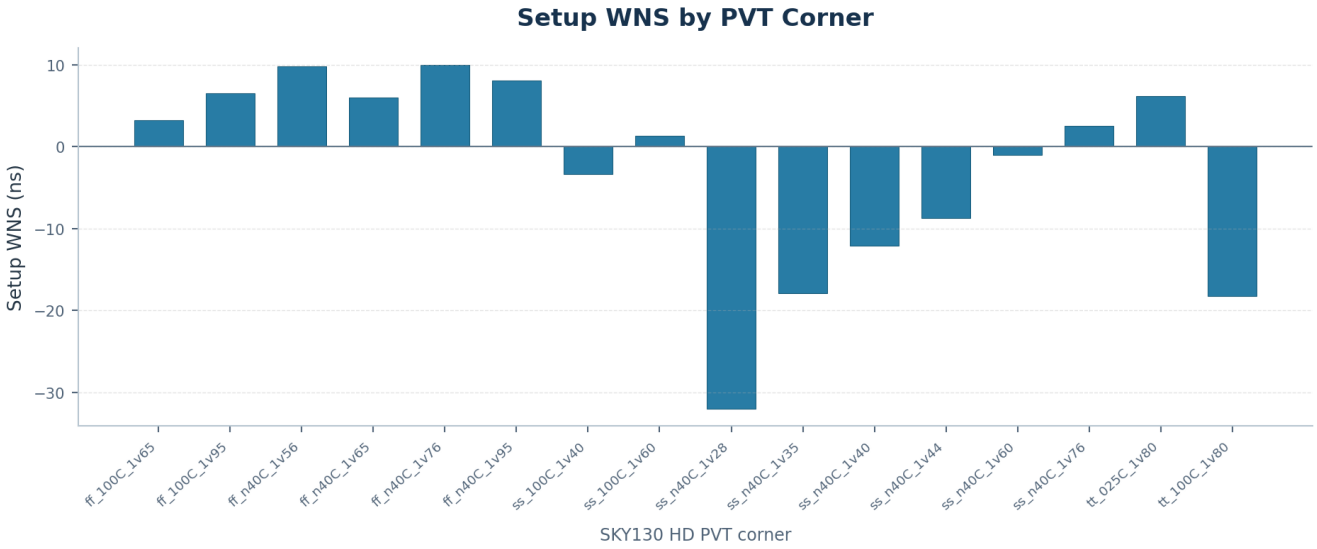
Timing metrics are in nanoseconds.

PVT Corner	P	Temp	V	Setup WNS	Setup TNS	Hold Slack	Hold TNS
ff_100C_1v65	ff	100	1.65	3.27	0.00	1.50	0.00
ff_100C_1v95	ff	100	1.95	6.50	0.00	3.25	0.00
ff_n40C_1v56	ff	-40	1.56	9.80	0.00	5.00	0.00
ff_n40C_1v65	ff	-40	1.65	6.00	0.00	0.26	0.00
ff_n40C_1v76	ff	-40	1.76	10.00	0.00	0.23	0.00
ff_n40C_1v95	ff	-40	1.95	8.06	0.00	0.19	0.00
ss_100C_1v40	ss	100	1.40	-3.34	-462.24	0.85	0.00
ss_100C_1v60	ss	100	1.60	1.29	0.00	0.64	0.00
ss_n40C_1v28	ss	-40	1.28	-32.03	-15425.60	2.75	0.00
ss_n40C_1v35	ss	-40	1.35	-17.90	-7390.62	1.23	0.00
ss_n40C_1v40	ss	-40	1.40	-12.08	-3942.03	3.10	0.00
ss_n40C_1v44	ss	-40	1.44	-8.72	-2114.08	0.95	0.00
ss_n40C_1v60	ss	-40	1.60	-1.00	-19.01	0.65	0.00
ss_n40C_1v76	ss	-40	1.76	2.52	0.00	1.85	0.00
tt_025C_1v80	tt	25	1.80	6.16	0.00	0.32	0.00
tt_100C_1v80	tt	100	1.80	-18.22	-5588.14	0.30	0.00

Summary: Seven rows have negative setup WNS; six have negative setup TNS. All supplied hold worst-slack values are positive and all hold TNS values are zero.

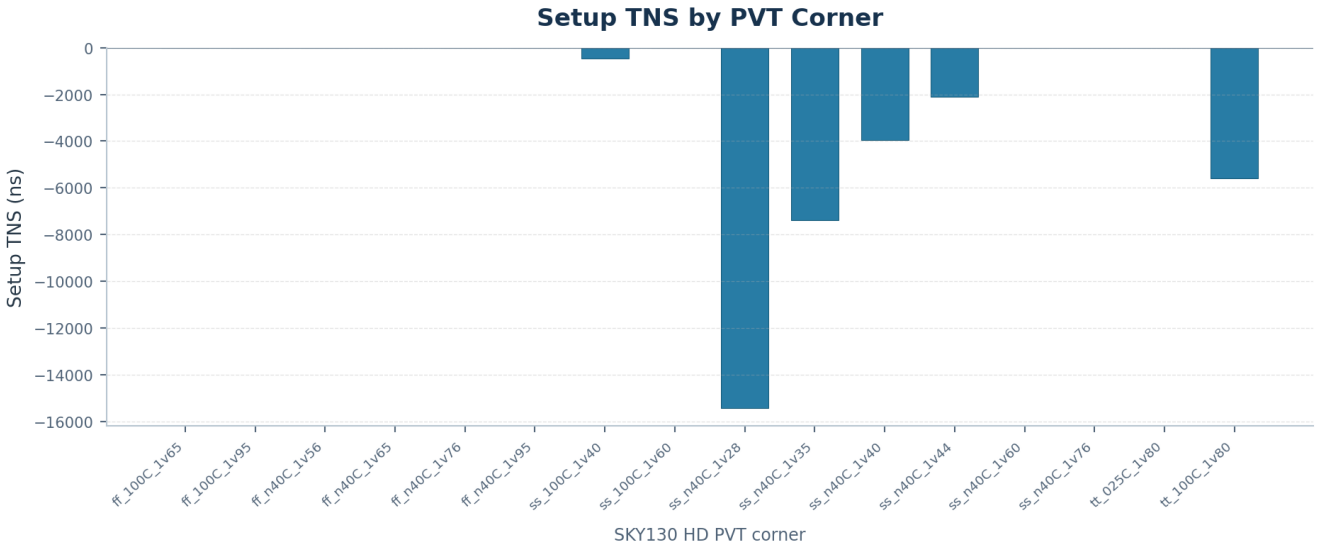
4. Setup Timing Analysis

4.1 Setup WNS by PVT Corner



Negative bars indicate setup timing violations. The minimum setup WNS is -32.03 ns at ss_n40C_1v28.

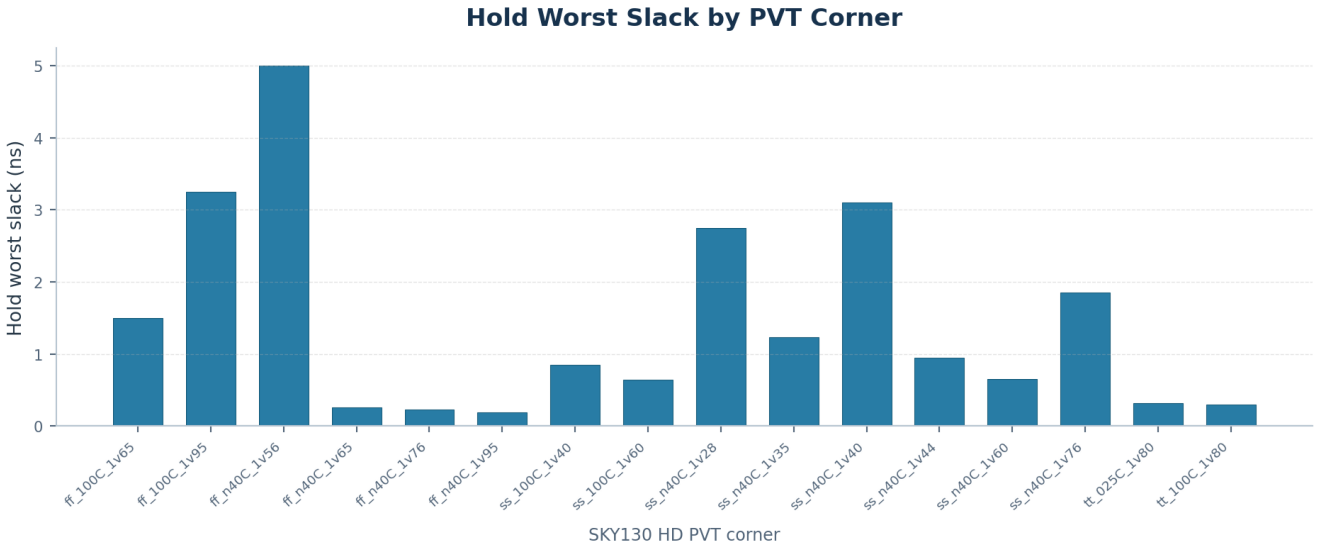
4.2 Setup TNS by PVT Corner



The most negative aggregate setup slack is -15425.60 ns at ss_n40C_1v28.

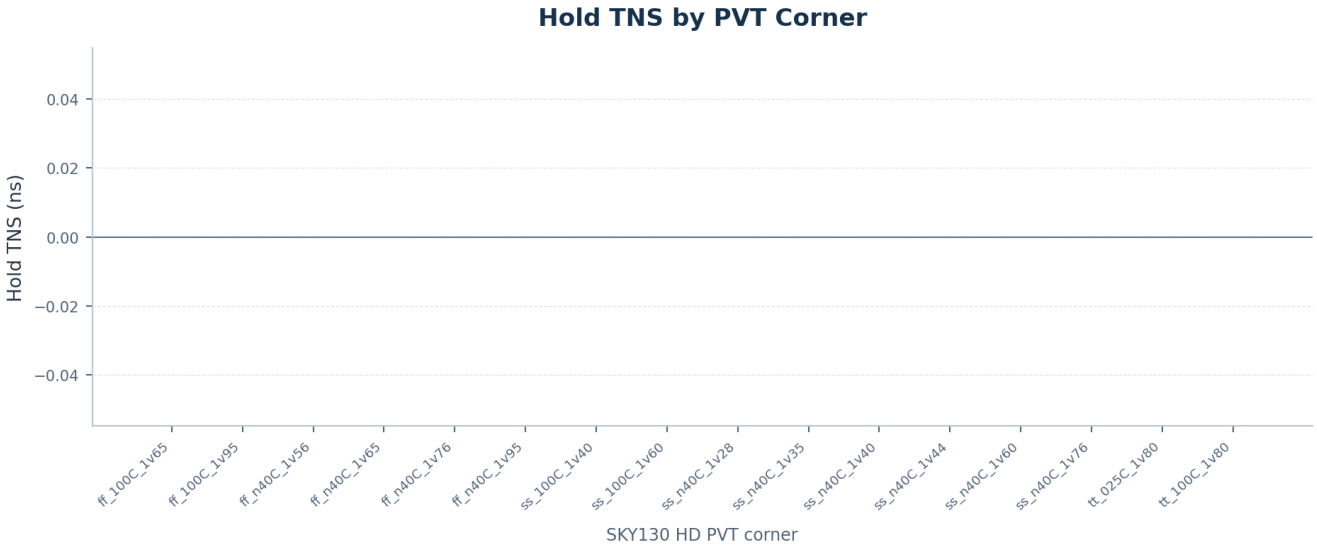
5. Hold Timing Analysis

5.1 Hold Worst Slack by PVT Corner



All supplied hold worst-slack values are positive. The smallest is 0.19 ns at ff_n40C_1v95.

5.2 Hold TNS by PVT Corner



All 16 hold TNS values are 0.00 ns; a flat zero line is the expected representation of these supplied values.

6. Critical Setup Path

The reference report identifies the worst setup path at ss_n40C_1v28.

Startpoint	core.CPU_src1_value_a3[14]$_DFF_P_
Endpoint	core.CPU_Xreg_value_a4[10][27]$_SDFFE_PP0P_
Data arrival	46.45 ns
Data required	14.42 ns
Slack	-32.03 ns

Slack = required time − arrival time = 14.42 ns − 46.45 ns = −32.03 ns. This confirms the reported violation magnitude; inspect the full path report, cell/net delay breakdown, clock arrival and uncertainty, path group, constraints, and exceptions to identify the root cause.

7. Physical-Design Evidence

Figure 1 — Floorplan / DAC macro view

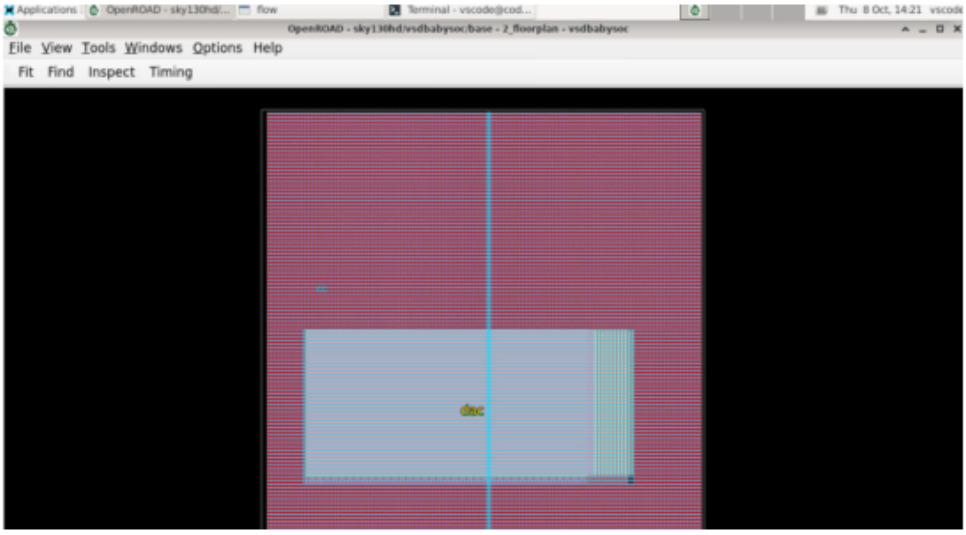


Figure 2 — Detailed routing around PLL macro



6. Physical-Design and Tool Evidence

Routed implementation and STA terminal summary

Figure 3 · Routed design overview

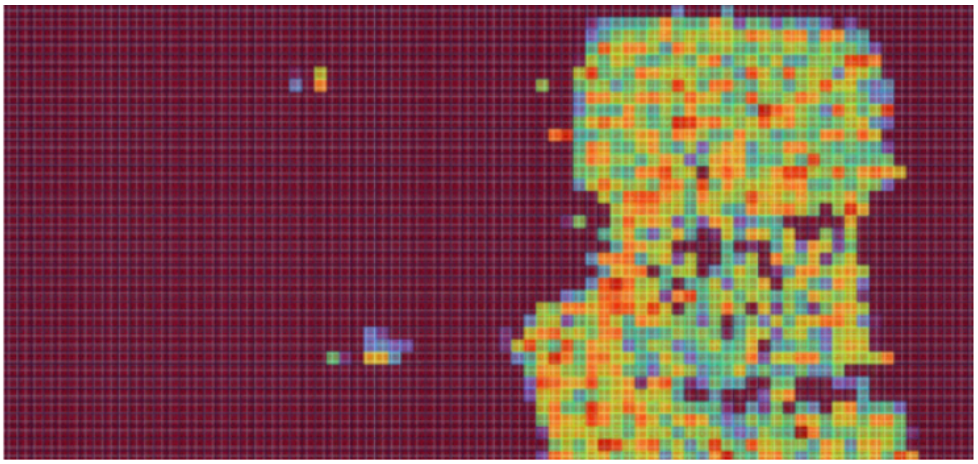
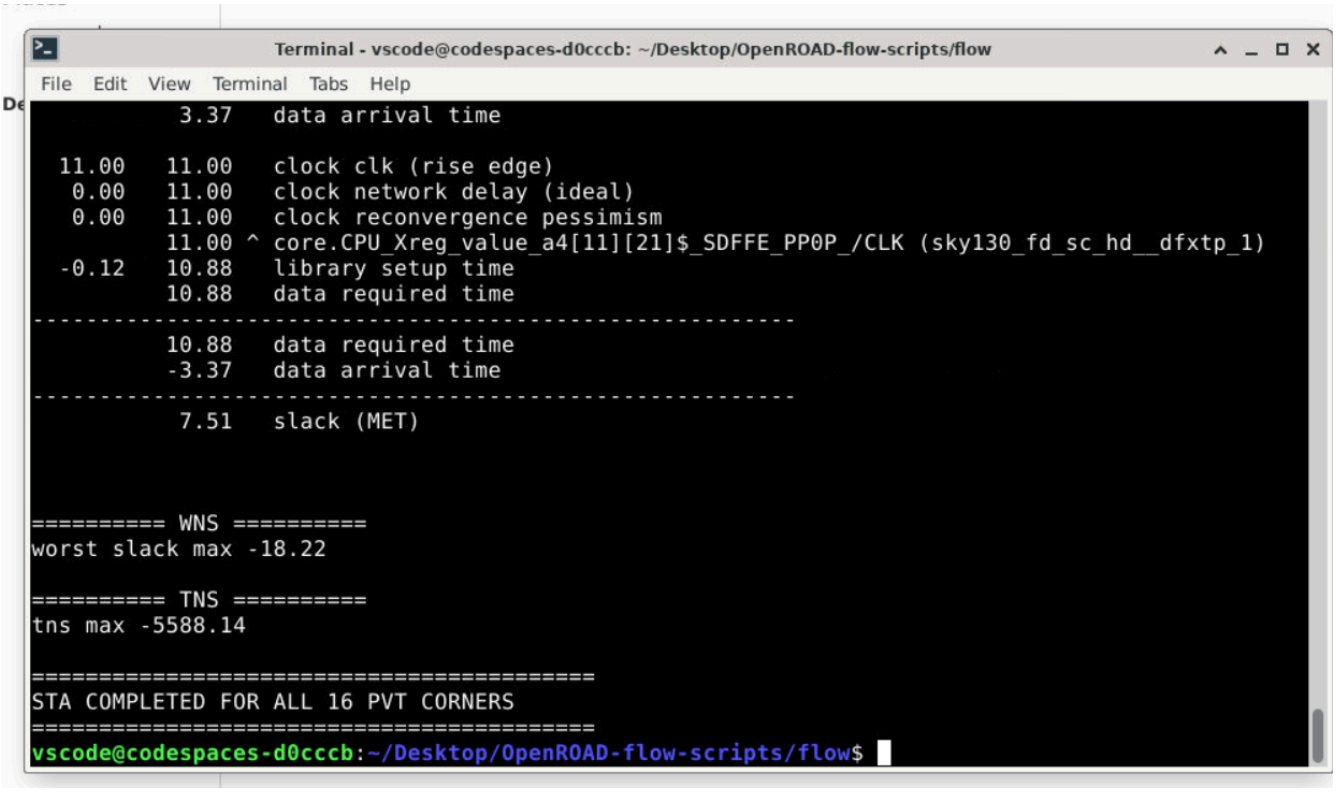
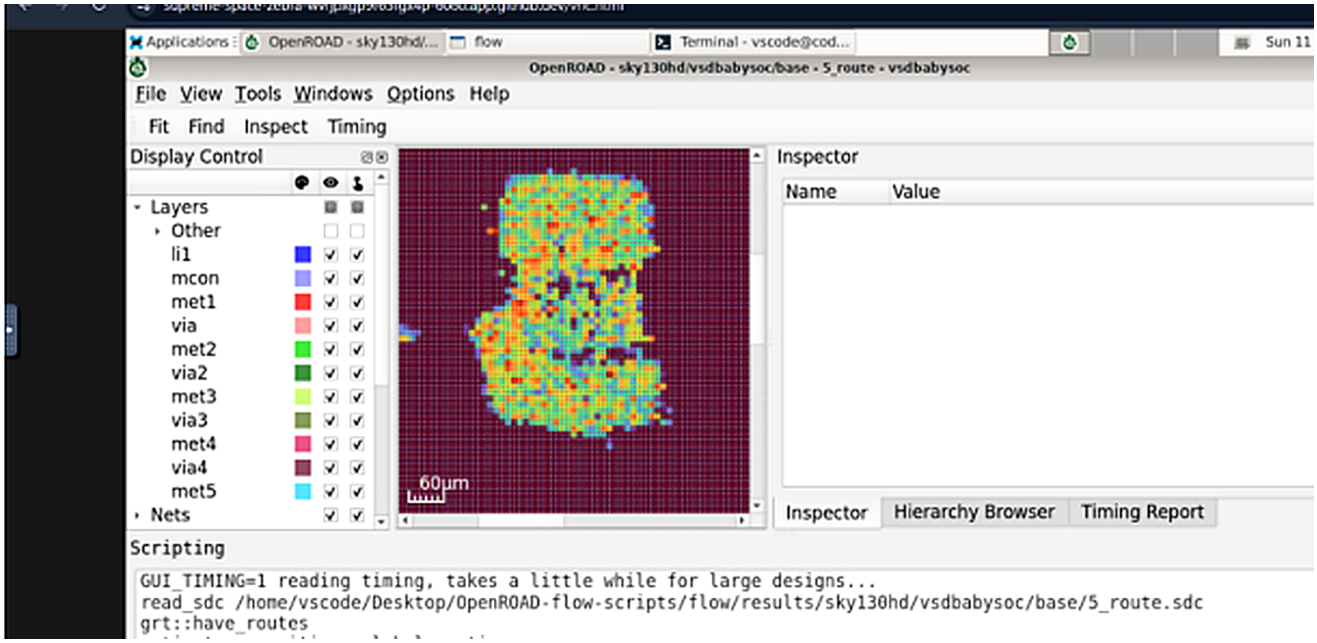


Figure 4 · STA terminal summary



8. Physical-Design Evidence (continued)

Figure 5 — OpenROAD GUI detailed view



What the figure shows

The OpenROAD GUI displays the routed design layout, visible routing layers, and the scripting/timing panel. This is supporting visual evidence of the implementation view; timing closure and physical verification must be established from the corresponding STA and DRC reports.

Image clarity improved by cropping away the browser chrome and enlarging the application view. Fine interface text remains limited by the resolution of the original screenshot.

9. Findings, Limitations and Next Steps

Findings

• Setup timing fails in multiple corners, especially ss_n40C_1v28, ss_n40C_1v35, ss_n40C_1v40 and ss_n40C_1v44.
• Hold worst slack is positive and hold TNS is zero in all supplied rows.
• The reference material notes 19 detailed-routing DRC entries categorized as “Short” near the DAC macro boundary; confirm these against the current DRC report.

Limitations

• The supplied timing table differs from some values in the earlier PDF; verify against original logs.
• The same SPEF across Liberty corners does not model a distinct RC extraction corner for each PVT condition.
• Available evidence does not establish full path coverage, constraint correctness or signoff readiness.

Recommended next steps

1. Inspect the complete worst setup path and cell/net delay breakdown.
2. Verify clock definitions, input/output delays, uncertainty, exceptions and path groups.
3. Use appropriate corner-specific parasitics where supported.
4. Investigate the 19 “Short” DRC entries near the DAC macro boundary.
5. Optimize, rerun STA and DRC, and archive scripts, logs, constraints, libraries and extraction provenance.

Conclusion

The 16-corner VSDBabySoC post-route STA comparison shows significant setup violations, with the most severe result at ss_n40C_1v28. The supplied hold metrics are positive/zero but do not offset setup failures. Further timing optimization and physical verification are required before signoff.

Prepared by Anirudh | 24EG104A59